计算机网络-实践课

Computer Networks

大作业

2025.12

实践课教务

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大作业.
SDN-Mininet

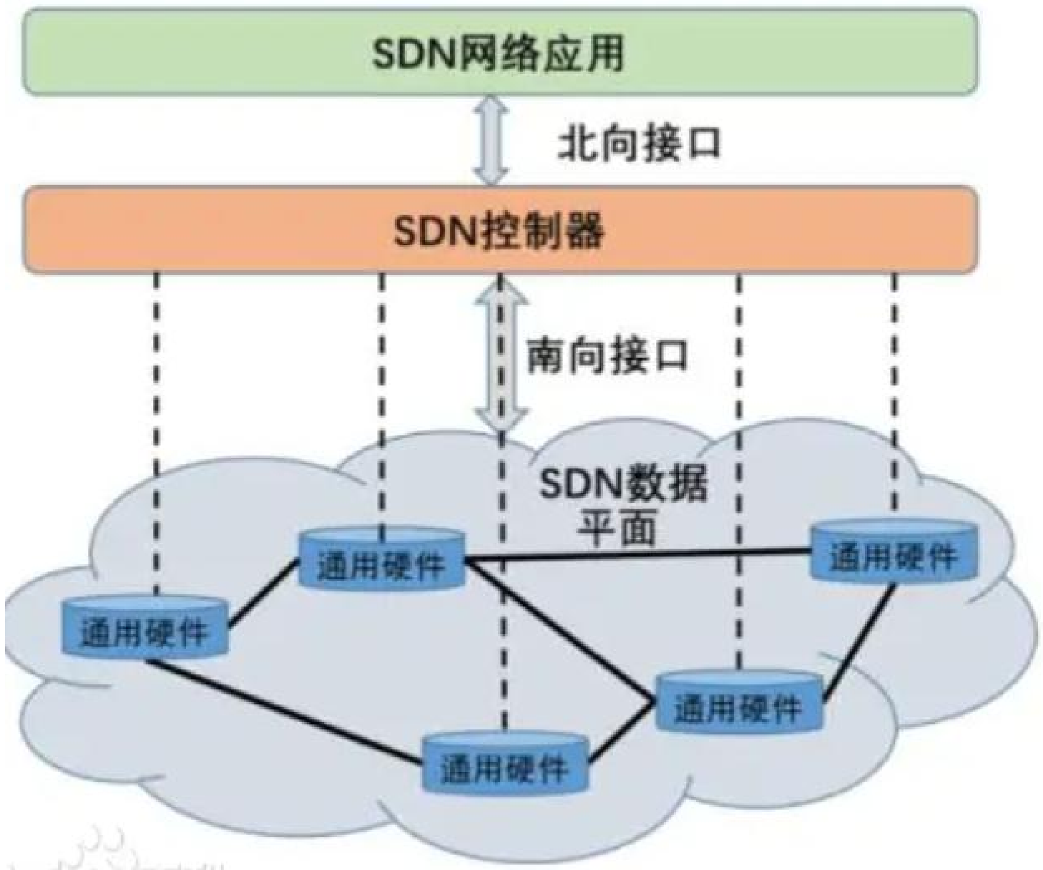
SDN (Software-Defined Networking)

- **软件定义网络（SDN: Software-Defined Networking）：**

□ 传统的网络架构中，网络设备（如**交换机**和**路由器**）通常内置有控制功能，设备自身可以根据算法和策略进行数据包转发和路由选择。

□ SDN提出一种新兴网络架构，采用集中式的**网络控制器**来对整个网络进行统一管理和配置，将网络控制平面（Control Plane）和数据转发平面（Data Plane）分离。**控制器充当网络的“大脑”**，并向交换机发送有关如何处理流量的命令。

SDN (*Software-Defined Networking*)



- 核心组件

☐ 控制器:

- ➢ 网络大脑，负责下发路由决策和策略
- ➢ *POX*、Floodlight、Ryu

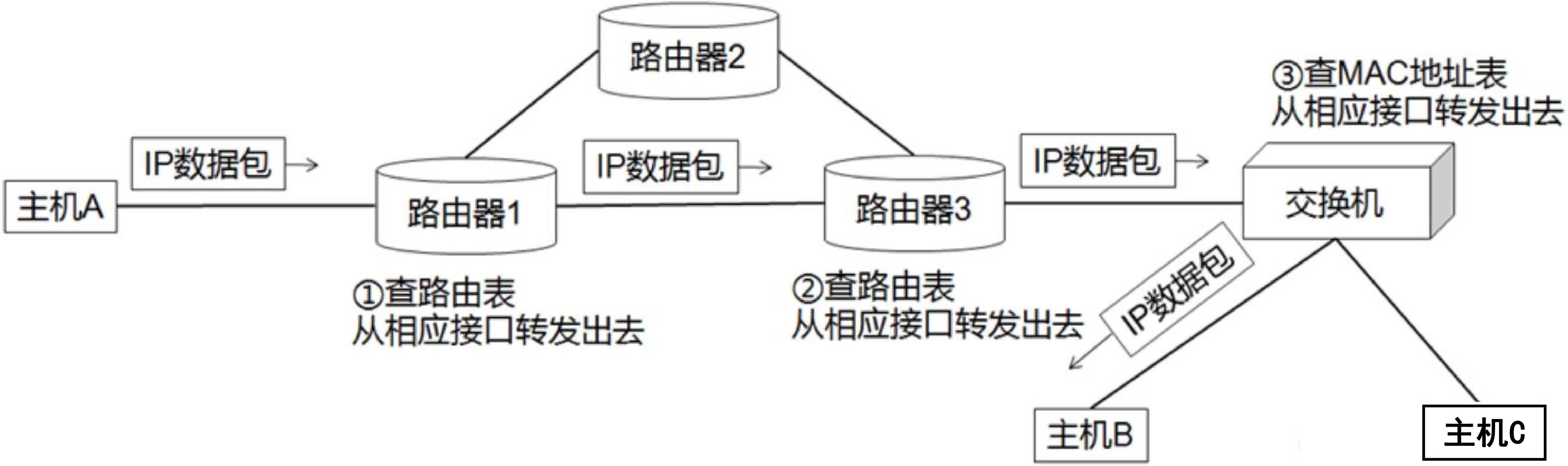
☐ 南向接口和南向协议:

- ➢ 定义控制器与网络设备(*switch, router*)的通信协议;
- ➢ *OpenFlow*、*OVSDB*

☐ 数据平面、北向接口/协议

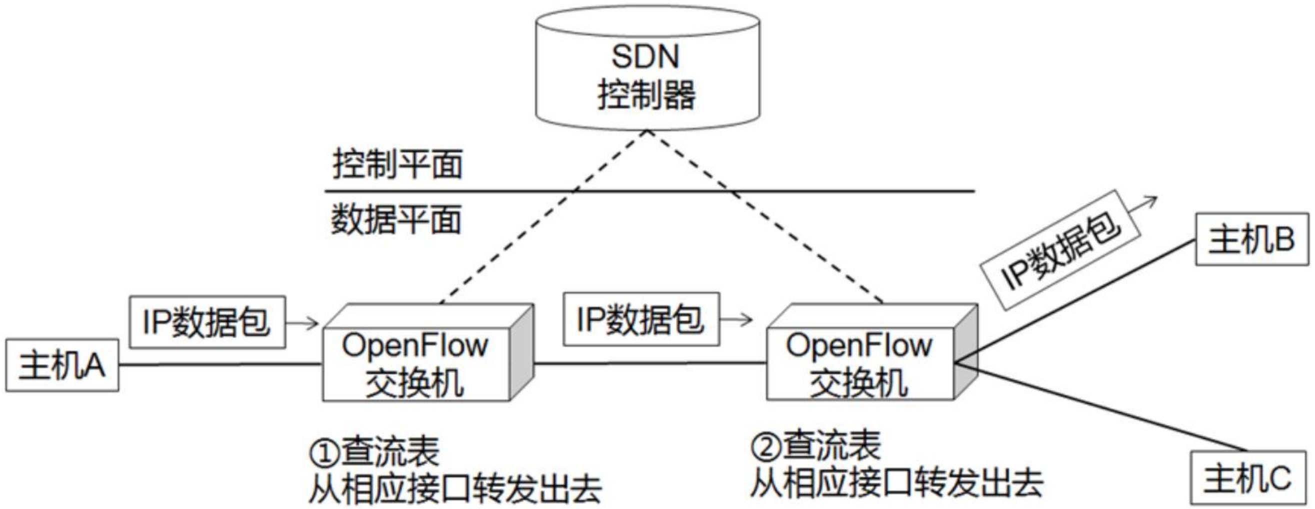
Openflow

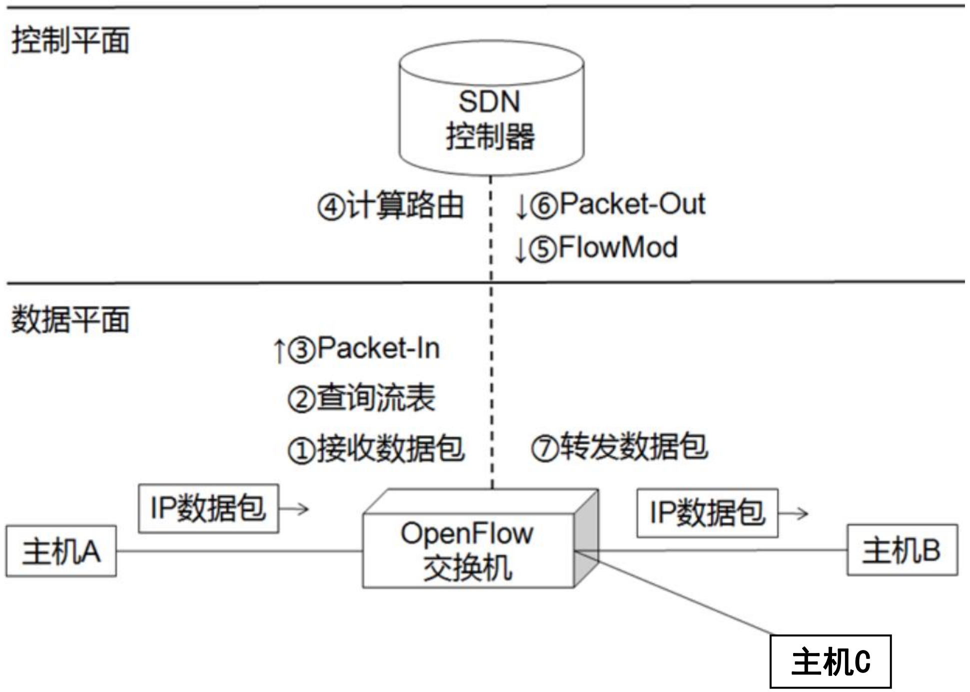
☐ 传统网络通信



Openflow

◻ SDN-OpenFlow通信





Openflow-流表（Flow Table）

FlowTable:

- 相当于*L2*链路层的*MAC*地址表和*L3*网络层的路由表

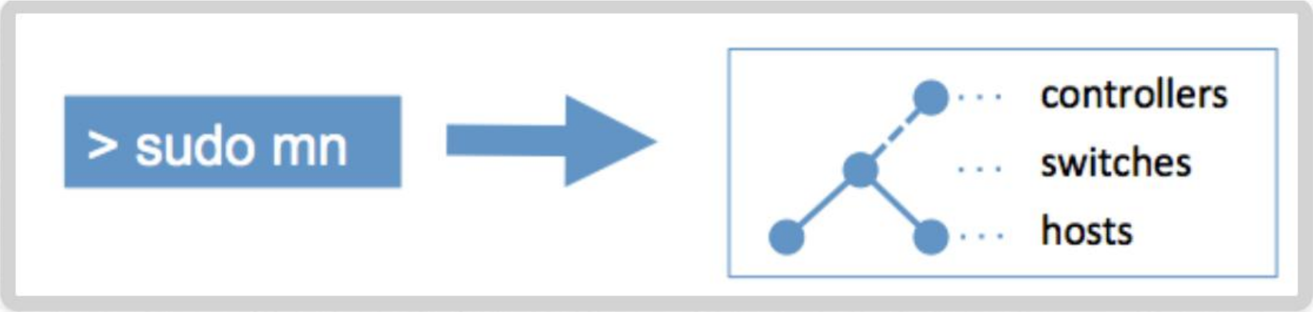
Flow Entry:

- 定义一条转发规则，由 *Match-Field* （匹配域）和 *instruction-Action*（动作）决定

```
OFPXMT_OFB_IP_DSCP      = 8,  /* IP DSCP (6 bits in ToS field). */
OFPXMT_OFB_IP_ECN       = 9,  /* IP ECN (2 bits in ToS field). */
OFPXMT_OFB_IP_PROTO     = 10, /* IP protocol. */
OFPXMT_OFB_IPV4_SRC     = 11, /* IPv4 source address. */
OFPXMT_OFB_IPV4_DST     = 12, /* IPv4 destination address. */
OFPXMT_OFB_TCP_SRC      = 13, /* TCP source port. */
OFPXMT_OFB_TCP_DST      = 14, /* TCP destination port. */
OFPXMT_OFB_UDP_SRC      = 15, /* UDP source port. */
OFPXMT_OFB_UDP_DST      = 16, /* UDP destination port. */
OFPXMT_OFB_SCTP_SRC     = 17, /* SCTP source port. */
OFPXMT_OFB_SCTP_DST     = 18, /* SCTP destination port. */
OFPXMT_OFB_ICMPV4_TYPE  = 19, /* ICMP type. */
OFPXMT_OFB_ICMPV4_CODE  = 20, /* ICMP code. */
OFPXMT_OFB_ARP_OP       = 21, /* ARP opcode. */
OFPXMT_OFB_ARP_SPA      = 22, /* ARP source IPv4 address. */
OFPXMT_OFB_ARP_TPA      = 23, /* ARP target IPv4 address. */
OFPXMT_OFB_ARP_SHA      = 24, /* ARP source hardware address. */
OFPXMT_OFB_ARP_THA      = 25, /* ARP target hardware address. */
```

- **output {port_no}**：将数据包转发到指定的 Port。
- **drop**：直接丢弃数据包。
- **set-field {field_type} {value}**：修改数据包头。

Mininet:



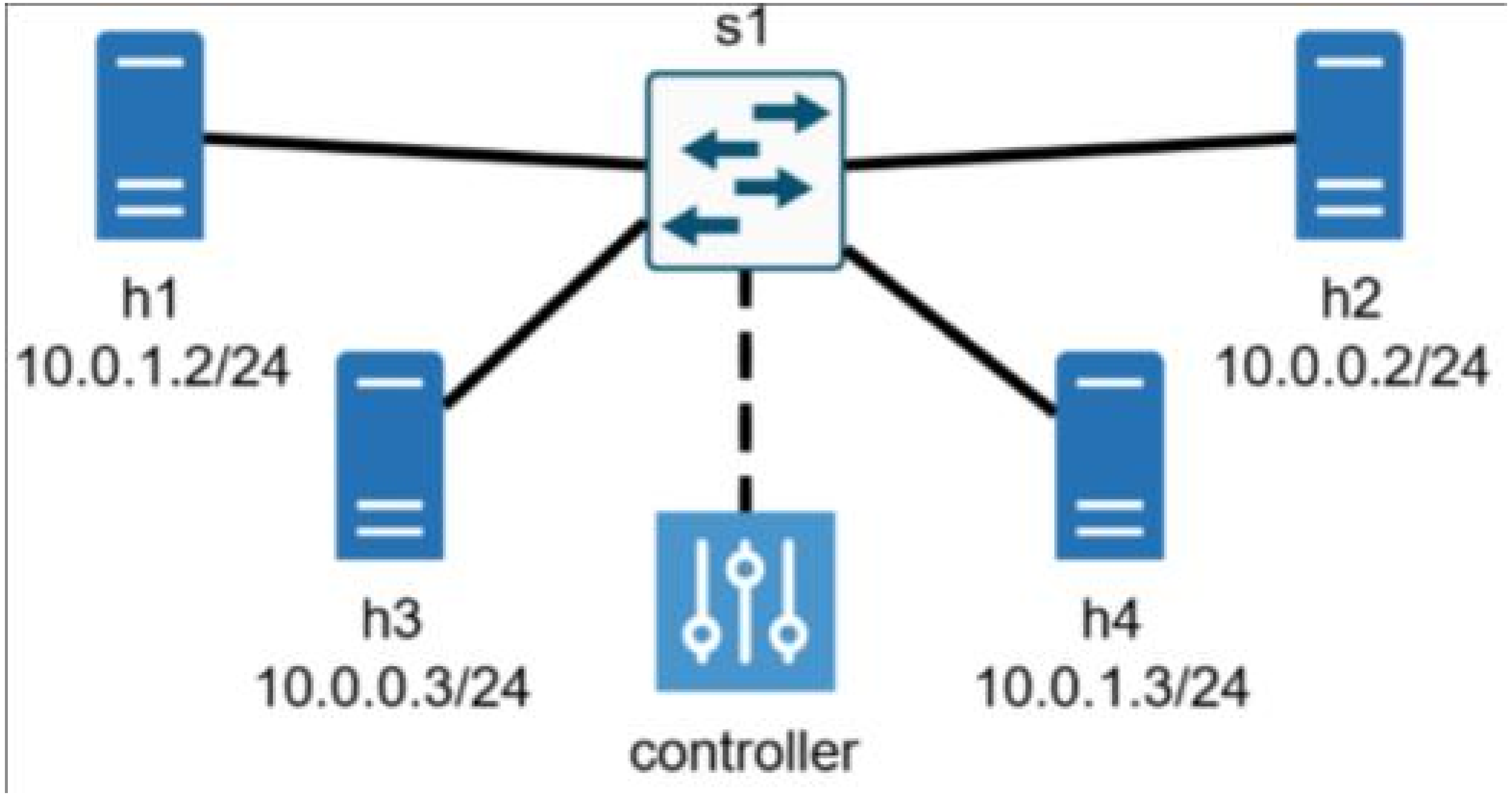
- 虚拟化网络仿真工具：创建包含主机、交换机、控制器的虚拟网络

- Mininet **CLI**: Simple **C**ommand-**L**ine **I**nterface to talk to nodes

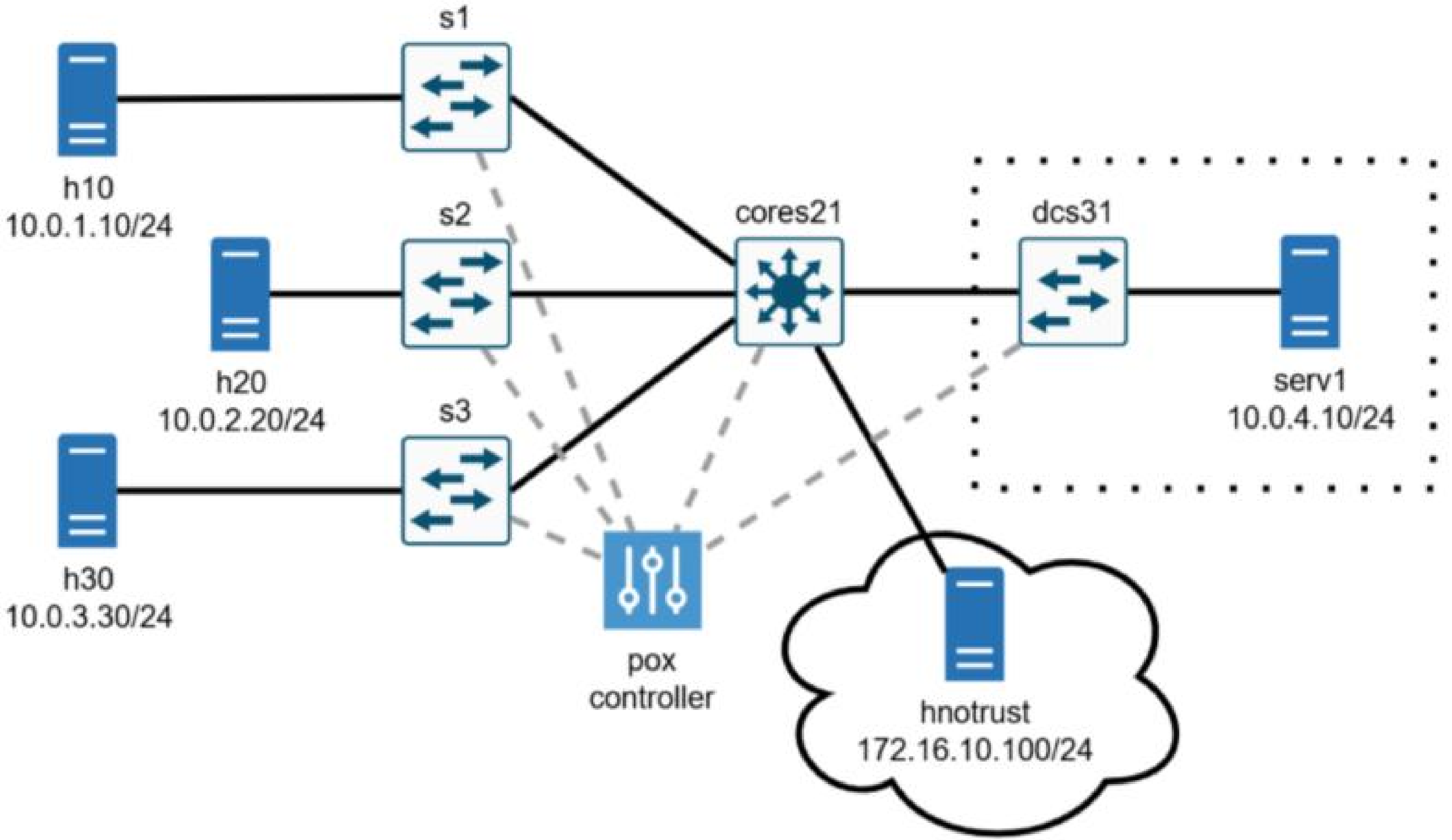
- **SDN实验和开发：** OpenFlow binaries and tools pre-installed

https://mininet.org/

Mininet: Part 1-2 Topology



Mininet: Part 3-4 Topology



Q&A Part 2:

Question-1: POX controller如何与switch进行通信?

Answer： connection.send(...)

如：connection.send(of.ofp_flow_mod) 可安装一条flow table entry

参考 https://github.com/mininet/openflow-tutorial/wiki/Create-a-Learning-Switch#user-content-Sending_OpenFlow_messages_with_POX

Question-2: Openflow 中，如何定义一个flow table entry?

Answer： fm = of.ofp_flow_mod(action=xxx， match=xxx， priority=xx)

参考 https://github.com/mininet/openflow-tutorial/wiki/Create-a-Learning-Switch#user-content-ofp_flow_mod_OpenFlow_message

Q&A Part 2:

Question-3: 如何定义flow table entry中的match和actions对象？

Answer：

Match：of.ofp_match()

`ofp_match` attributes:

Attribute	Meaning
in_port	Switch port number the packet arrived on
dl_src	Ethernet source address
dl_dst	Ethernet destination address
dl_vlan	VLAN ID
dl_vlan_pcp	VLAN priority
dl_type	Ethertype / length (e.g. 0x0800 = IPv4)
nw_tos	IP TOS/DS bits
nw_proto	IP protocol (e.g., 6 = TCP) or lower 8 bits of ARP opcode
nw_src	IP source address
nw_dst	IP destination address
tp_src	TCP/UDP source port
tp_dst	TCP/UDP destination port

参考 https://noxrepo.github.io/pox-doc/html/#match-structure

Q&A Part 2:

Question-3: 如何定义flow table entry中的match和actions对象？

Answer：

转发动作:

of.ofp_action_output()

Structure definition:

```
class ofp_action_output (object):
  def __init__ (self, **kw):
    self.port = None # Purposely bad -- require specification
```

- port (int) the output port for this packet. Value could be an actual port number or one of the following virtual ports:
  - OFPP_IN_PORT - Send back out the port the packet was received on. Except possibly OFPP_NORMAL, *this is the only way to send a packet back out its incoming port*.
  - OFPP_TABLE - Perform actions specified in flowtable. Note: Only applies to ofp_packet_out messages.
  - OFPP_NORMAL - Process via normal L2/L3 legacy switch configuration (if available – switch dependent)
  - OFPP_FLOOD - output all openflow ports except the input port and those with flooding disabled via the OFPPC_NO_FLOOD port config bit (generally, this is done for STP)
  - OFPP_ALL - output all openflow ports except the in port.
  - OFPP_CONTROLLER - Send to the controller.
  - OFPP_LOCAL - Output to local openflow port.
  - OFPP_NONE - Output to no where.

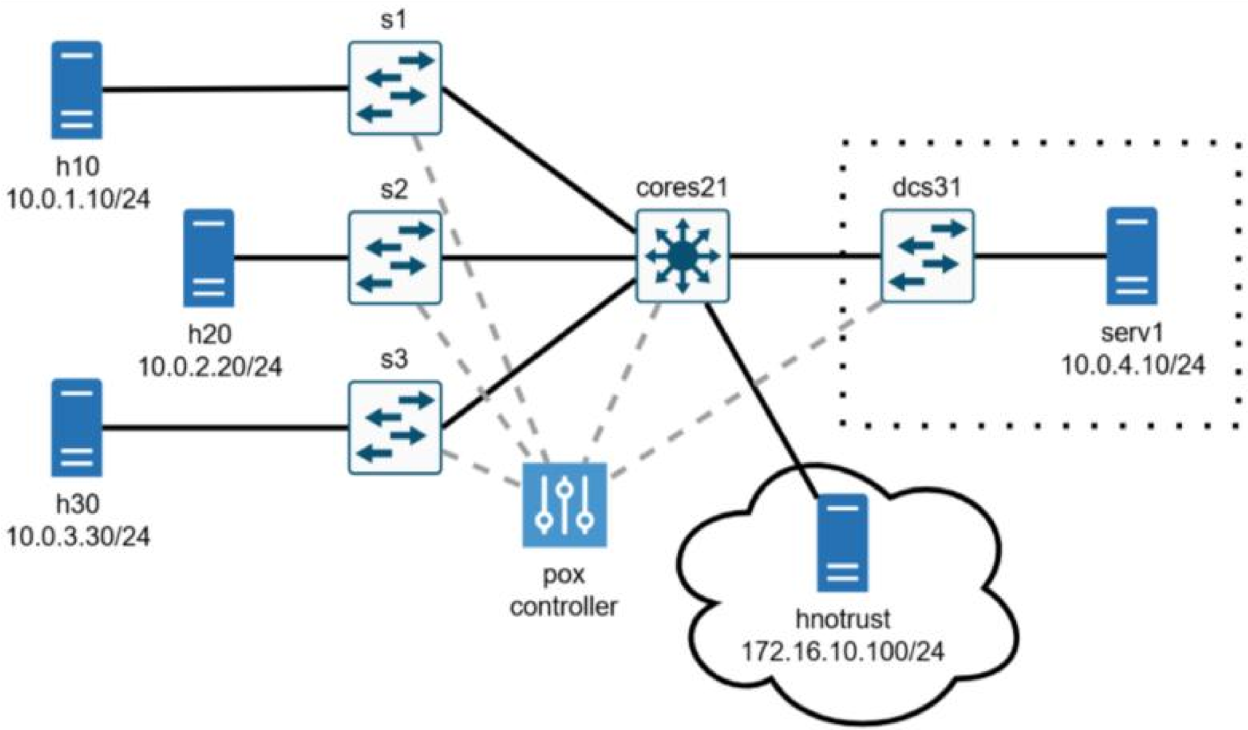
参考

https://noxrepo.github.io/pox-doc/html/#openflow-actions

Q&A Part 3：

Question-1：网络结构定义，及静态ARP表？

Answer：topos/part1.py 已给出，且提前为所有主机都存储了全局ARP表




```
# Convenience mappings of hostnames to ips
IPS = {
    "h10": "10.0.1.10",
    "h20": "10.0.2.20",
    "h30": "10.0.3.30",
    "serv1": "10.0.4.10",
    "hnotrust": "172.16.10.100",
}

# Convenience mappings of hostnames to subnets
SUBNETS = {
    "h10": "10.0.1.0/24",
    "h20": "10.0.2.0/24",
    "h30": "10.0.3.0/24",
    "serv1": "10.0.4.0/24",
    "hnotrust": "172.16.10.0/24",
}
```

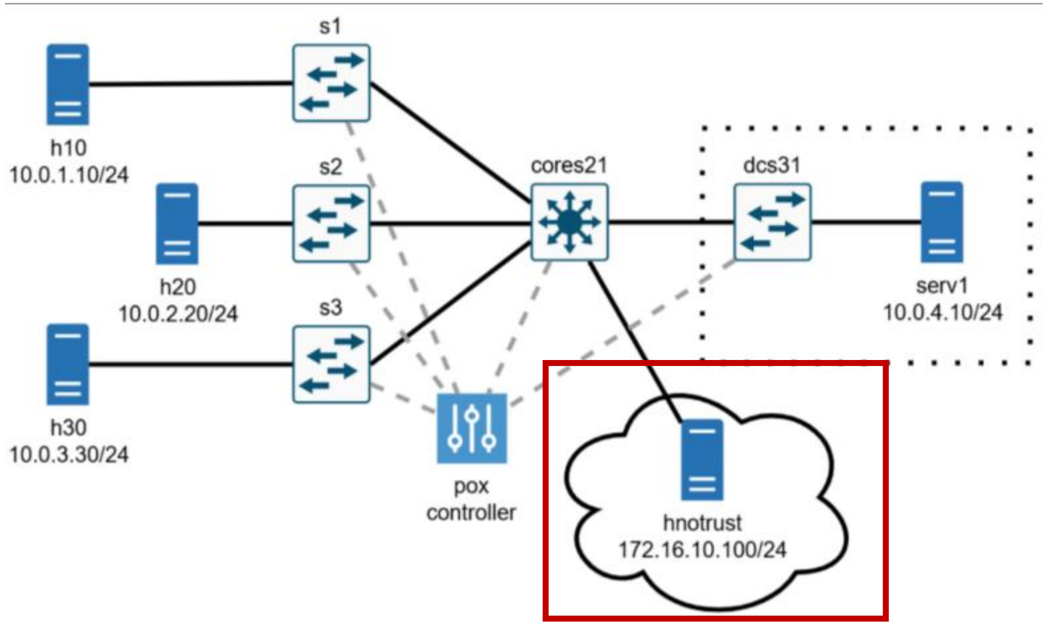
```
# Pre-populate all neighbor tables
hosts = ["h10", "h20", "h30", "serv1", "hnotrust1"]
for configuring_host_name in hosts:
    configuring_host = net.get(configuring_host_name)
    for remote_host_name in hosts:
        if configuring_host_name != remote_host_name:
            remote_host = net.get(remote_host_name)
            configuring_host.setARP(remote_host.IP(), remote_host.MAC()
```

Q&A Part 3：

Question-2：hnotrust的packet在哪个switch上处理？

Answer：

Core21上，即修改core21_setup()，阻拦来自hnotrust的全部ICMP包 以及 从hnotrust到serv1的IP包



Q&A Part 4：

Question-1: TODO?
Answer：
Part 4没有为各主机配置全局ARP表，且各主机将数据包发送给子网网关

修改_handle_PacketIn()：无法正常路由的包会转发到此函数
如何解析收到包，涉及event和packet处理，见参考

增加了部分TODO说明和代码引导，见a2part2controller.py文件
参考：https://noxrepo.github.io/pox-doc/html/#packetin
https://noxrepo.github.io/pox-doc/html/#working-with-packets-pox-lib-packet

Q&A Part 4：

Question-2: 如何生成ARP reply 并 转发?

Answer：

可使用pox.lib.packet库

参考： https://noxrepo.github.io/pox-doc/html/#working-with-packets-pox-lib-packet